

Data Structures (SOC2010)

Submission Deadline: 18/09/ 2023 (16:45 Hr.) Monday

NOTE:

1. You are required to **take print out of this sheet** and solve (**handwritten**) the questions on sheet only.
[Submissions not conforming to the given format will not be evaluated]
2. **Copying** of solution is strictly prohibited. In case it is found, then zero marks will be awarded.
3. Submit the scan copy of your solution on eclass. Late submissions are not accepted.

Questions

1. What is the running time of the program? (Write explanation for your answer)

```
void fun(int n, int arr[])
{
    int i = 0, j = 0;
    for(; i < n; ++i)
        while(j < n && arr[i] < arr[j])
            j++;
}
```

Answer _____

2. Bubble sort is used to sort the series. Each pass (iteration) of bubble sort will work as follows:
Take an array of numbers " **5 1 4 2 8**", after first pass the sequence will be "**1 4 2 5 8**"

Explanation as follows:

Sequence of First Pass can be found as follows

(**5 1 4 2 8**) → (**1 5 4 2 8**), compares the first two elements, and swaps since $5 > 1$.

(**1 5 4 2 8**) → (**1 4 5 2 8**), Swap since $5 > 4$

(**1 4 5 2 8**) → (**1 4 2 5 8**), Swap since $5 > 2$

(**1 4 2 5 8**) → (**1 4 2 5 8**), Now, since these elements are already in order ($8 > 5$).

Apply Bubble sort on the following sequence to sort and **write only the state of an array** after each pass

(*make sure that algorithms behaves in adaptive manner*)

Array	3	5	8	4	1	6	12	9
1st Pass								
2 nd Pass								
3 rd Pass								
4 th Pass								
5 th Pass								
6 th Pass								
7 th Pass								

3. Apply **selection sort** on the following sequence to arrange the numbers in ascending order. Also comment on its running time.

10	17	6	2	20

Running Time = _____